

Missings

```
df_subset$missing_per_row <- rowSums(is.na(df_subset))

missing_row_table <- df_subset %>%
  count(missing_per_row, name = "num_rows") %>%
  arrange(missing_per_row)

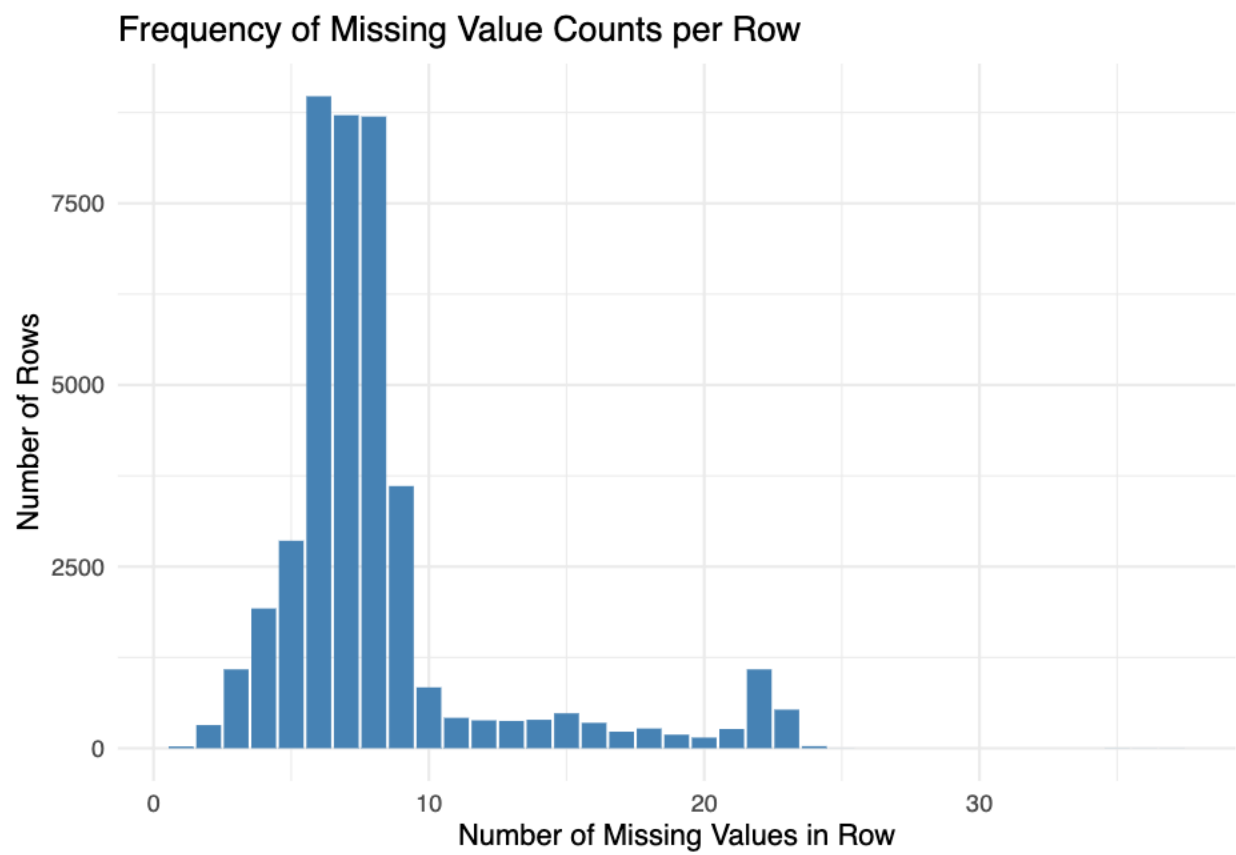
kable(missing_row_table, caption = "Number of Rows by Count of Missing Values")
```

Table 1: Number of Rows by Count of Missing Values

missing_per_row	num_rows
1	25
2	319
3	1086
4	1924
5	2856
6	8973
7	8712
8	8694
9	3609
10	838
11	419
12	384
13	376
14	392
15	481
16	350
17	228
18	272
19	187
20	147
21	265
22	1086
23	532
24	29
25	1
35	2
36	1
37	1

```
df_subset %>%
  count(missing_per_row) %>%
  ggplot(aes(x = missing_per_row, y = n)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Frequency of Missing Value Counts per Row",
    x = "Number of Missing Values in Row",
    y = "Number of Rows"
```

```
) +  
theme_minimal()
```



```

missing_per_column <- colSums(is.na(df_subset))

missing_table <- tibble(
  variable = names(missing_per_column),
  num_missing = as.integer(missing_per_column),
  pct_missing = round(100 * missing_per_column / nrow(df_subset), 2)
) %>%
  arrange(desc(num_missing))

kable(missing_table, caption = "Missing Values per Variable in df_subset")

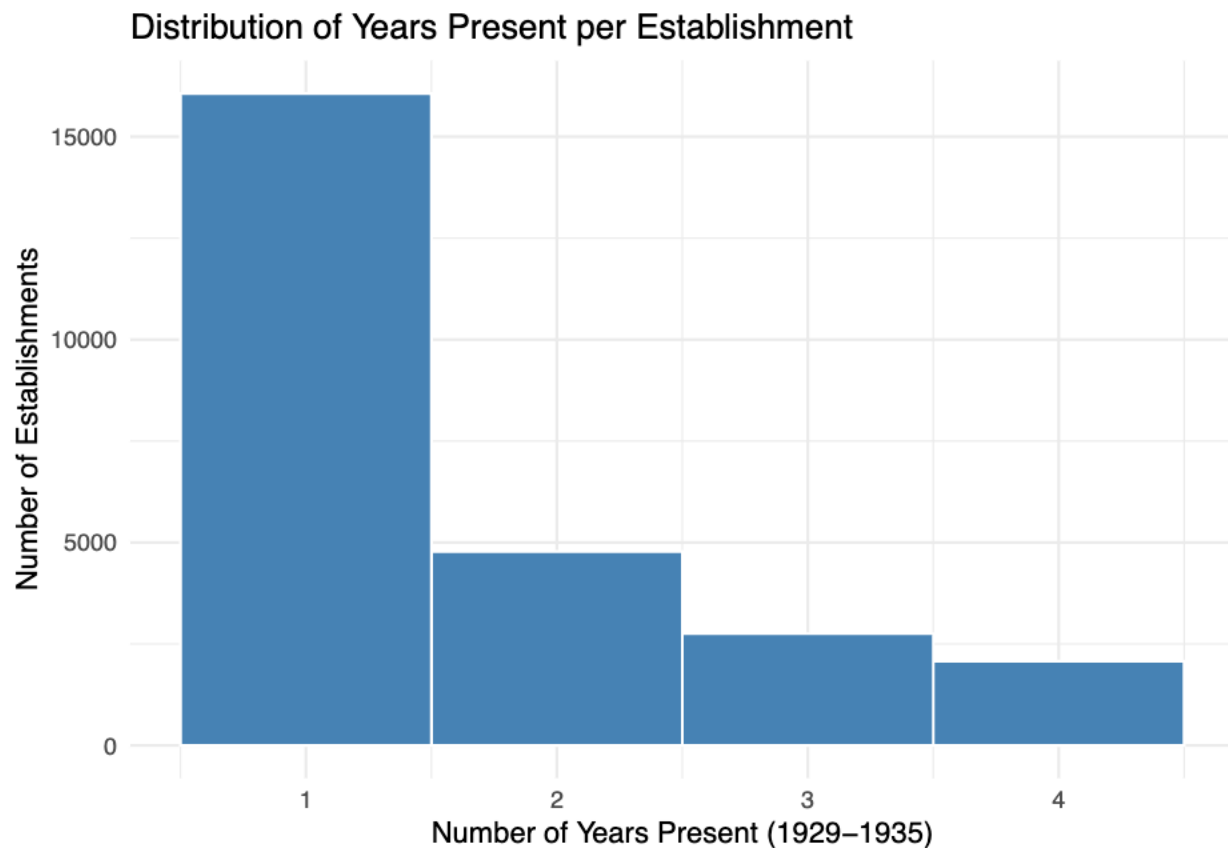
```

Table 2: Missing Values per Variable in df_subset

variable	num_missing	pct_missing
business_change_as004	39154	92.81
former_info_as005	37688	89.33
ownership_change_as003	37055	87.83
location_change_as002	36537	86.60
name_change_as001	35662	84.53
materials_fuel_contract_cost_f007	34207	81.08
total_input_cost_f008	31630	74.97
general_office_address_a004	17025	40.35
street_and_number_ag004	10653	25.25
owner_name_a002	6064	14.37
wage_earners_jan_ewemt01	5458	12.94
wage_earners_feb_ewemt02	5361	12.71
wage_earners_nov_ewemt11	4621	10.95
wage_earners_mar_ewemt03	4571	10.83
wage_earners_dec_ewemt12	4520	10.71
wage_earners_apr_ewemt04	4382	10.39
wage_earners_oct_ewemt10	4090	9.69
wage_earners_may_ewemt05	3928	9.31
wage_earners_jul_ewemt07	3536	8.38
wage_earners_aug_ewemt08	3517	8.34
wage_earners_sep_ewemt09	3177	7.53
wage_earners_jun_ewemt06	3142	7.45
incorporated_a003	2404	5.70
monthly_wage_earners_summary_ewemt	1826	4.33
plant_name_a001	505	1.20
place_ag002	261	0.62
total_wages_e005s	139	0.33
value_of_product_g000v	134	0.32
county_ag003	59	0.14
survey_year	11	0.03
firm_id	10	0.02
firm_id_v1	9	0.02
establishment_id	6	0.01
establishment_id_v1	6	0.01
sic_code	5	0.01
region	5	0.01
state_ag001	4	0.01

variable	num_missing	pct_missing
missing_per_row	0	0.00

```
df_subset %>%
  group_by(establishment_id) %>%
  summarise(years_present = n_distinct(.data$survey_year)) %>%
  ggplot(aes(x = years_present)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  scale_x_continuous(breaks = 1:4) +
  labs(
    title = "Distribution of Years Present per Establishment",
    x = "Number of Years Present (1929–1935)",
    y = "Number of Establishments"
  ) +
  theme_minimal()
```



```
df_subset %>%
  group_by(establishment_id) %>%
  summarise(years_present = n_distinct(.data$survey_year)) %>%
  count(years_present, name = "num_establishments") %>%
  mutate(pct = round(100 * num_establishments / sum(num_establishments), 1)) %>%
  kable(
    caption = "Distribution of Establishments by Number of Years Present",
    col.names = c("Years Present", "Number of Establishments", "Percent")
  )
```

)

Table 3: Distribution of Establishments by Number of Years Present

Years Present	Number of Establishments	Percent
1	16066	62.6
2	4779	18.6
3	2752	10.7
4	2076	8.1